

ParameterDict#

<https://pytorch.org/docs/stable/generated/torch.nn.ParameterDict.html>

Description:

Holds parameters in a dictionary. ParameterDict can be indexed like a regular Python dictionary, but Parameters it contains are properly registered, and will be visible by all Module methods. Other objects are treated as would be done by a regular Python dictionary. ParameterDict is an ordered dictionary. `update()` with other unordered mapping types (e.g., Python's plain dict) does not preserve the order of the merged mapping. On the other hand, `OrderedDict` or another `ParameterDict` will preserve their ordering. Note that the constructor, assigning an element of the dictionary and the `update()` method will convert any Tensor into Parameter. `values(iterable, optional)` – a mapping (dictionary) of (string : Any) or an iterable of key-value pairs of type (string, Any)

Function Signature

```
class torch.nn.ParameterDict(parameters=None)[source]#  
Parameters  
clear()[source]#  
copy()[source]#  
Return type
```

Returns:

Iterable[Any]

Code Examples

```
class MyModule(nn.Module):
    def __init__(self) -> None:
        super().__init__()
        self.params = nn.ParameterDict(
            {
                "left": nn.Parameter(torch.randn(5, 10)),
                "right": nn.Parameter(torch.randn(5, 10)),
            }
        )

    def forward(self, x, choice):
        x = self.params[choice].mm(x)
        return x
```

Notes & Warnings

NOTE If parameters is an OrderedDict, a ParameterDict, or an iterable of key-value pairs, the order of new elements in it is preserved.

Detailed Documentation

Documentation

Holds parameters in a dictionary.

ParameterDict can be indexed like a regular Python dictionary, but Parameters it contains are properly registered, and will be visible by all Module methods. Other objects are treated as would be done by a regular Python dictionary

ParameterDict is an ordered dictionary. update() with other unordered mapping types (e.g., Python's plain dict) does not preserve the order of the merged mapping. On the other hand, OrderedDict or another ParameterDict will preserve their ordering.

Note that the constructor, assigning an element of the dictionary and the update() method will convert any Tensor into Parameter.

values (iterable, optional) – a mapping (dictionary) of (string : Any) or an iterable of key-value pairs of type (string, Any)

Remove all items from the ParameterDict.

Return a copy of this ParameterDict instance.

Return a new ParameterDict with the keys provided.

keys (iterable, string) – keys to make the new ParameterDict from

default (Parameter, optional) – value to set for all keys

Return the parameter associated with key if present. Otherwise return default if provided, None if not.

key (str) – key to get from the ParameterDict

default (Parameter, optional) – value to return if key not present

Return an iterable of the ParameterDict key/value pairs.

Iterable[tuple[str, Any]]

Return an iterable of the ParameterDict keys.

Remove key from the ParameterDict and return its parameter.

key (str) – key to pop from the ParameterDict

Remove and return the last inserted (key, parameter) pair from the ParameterDict.

Set the default for a key in the Parameterdict.

If key is in the ParameterDict, return its value. If not, insert key with a parameter default and return default. default defaults to None.

key (str) – key to set default for

default (Any) – the parameter set to the key

Update the ParameterDict with key-value pairs from parameters, overwriting existing keys.

If parameters is an OrderedDict, a ParameterDict, or an iterable of key-value pairs, the order of new elements in it is preserved.

parameters (iterable) – a mapping (dictionary) from string to Parameter, or an iterable of key-value pairs of type (string, Parameter)

Return an iterable of the ParameterDict values.